Dr. Siqi Huang

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Pure Profile Google Scholar Scopus ORCID GitHub

ABOUT ME / RESEARCH PROFILE

Dr. Siqi Huang is an Assistant Professor in the Academy of Artificial Intelligence and Advanced Technology (AIAT) at Xi'an Jiaotong-Liverpool University (XJTLU), Entrepreneur College (Taicang). He received his Ph.D. in Electrical and Computer Engineering from the University of North Carolina at Charlotte in 2022, and his B.Eng. in Software Engineering from Sun Yat-sen University in 2015. His primary research interests encompass Edge Computing with Embedded AI Devices (TinyML), Computer Vision and Deep Learning, Vision LLMs & AIGC, Mobile Augmented Reality (MAR), and Autonomous Driving Systems. He serves as Programme Director for BEng Data Science and Big Data Technology with Contemporary Entrepreneurialism, and heads the XJTLU Metaverse Research Lab.

EDUCATION & PROFESSIONAL QUALIFICATIONS

Xi'an Jiaotong-Liverpool University, Suzhou, China Sep 2023 – Dec 2025
Postgraduate Certificate in Higher Education (PGCert) Modules

- Completed higher education pedagogical training modules in curriculum design, student assessment, and research-led teaching.

University of North Carolina at Charlotte, Charlotte, NC, USA Jan 2016 – May 2022
Ph.D. in Electrical and Computer Engineering

- Dissertation: *Analysis and Enhancement of Resource-hungry Applications* (Advisor: Prof. Linda Xie).

Sun Yat-sen University, Guangzhou, China Sep 2011 – July 2015
B.Eng. in Software Engineering

- Award: Software Engineering School Honored Student.

PROFESSIONAL EXPERIENCE

Xi'an Jiaotong-Liverpool University, Suzhou, China Aug 2022 – Present
Assistant Professor, School of AI and Advanced Computing

Key Responsibilities:

- Serve as Programme Director for the undergraduate BEng Data Science and Big Data Technology with Contemporary Entrepreneurialism (DSBD) programme, overseeing academic quality, curriculum review, Liverpool exchange mapping, and annual monitoring.
- Lead undergraduate modules (DTS101TC Introduction to Neural Networks, DTS311TC Final Year Project) and co-teach foundational and advanced computing modules (DTS002TC, DTS102TC, DTS106TC, DTS110TC, DTS203TC, DTS205TC, DTS207TC, ENT001).
- Direct the AIAC Metaverse Research Lab, mentoring undergraduate researchers, postgraduate students, and SURF teams on applied AI projects.
- Supervise doctoral research, JITRI Joint Master's dissertations, and Final Year Projects (34+ students completed/ongoing).
- Execute administrative duties as AIAC Student Recruitment and Outreach Officer, AIAC Pathfinder with EEH, and University Research Representative.

Key Achievements:

- Awarded Second Prize of Annual Curriculum Innovation at the First XJTLU Innovative Cases of Industry-Education Integrated Curriculum Competition (2024).

- Secured over RMB 11.2 million in competitive external research grants, enterprise joint lab funding, and governmental projects as PI and Co-PI.
- Published 7+ high-impact research papers in premier venues including IJCAI-2024 (CORE A*, CCF A), IEEE Transactions on Cloud Computing (SCI Q1), and ITU Journal on FET; secured 1 authorized patent and 1 pending patent.
- Mentored student competition teams to national and regional awards (Global Campus AI Elite Competition, National Smart Car Competition, CP Cup, Global Environment Challenge).
- Rated EXCELLENT (AY2022/23), OUTSTANDING (AY2023/24), and OUTSTANDING (AY2024/25) in annual Professional Development Reviews by Line Manager.

Modules Taught at XJTLU:

- **DTS002TC** Essentials of Big Data (Spring 2024, Spring 2025, Spring 2026)
- **DTS101TC** Introduction to Neural Networks (Spring 2025, Spring 2026 – Module Leader)
- **DTS102TC** Programming with C++ / C++ & R (Fall 2022, Fall 2024, Fall 2025)
- **DTS106TC** Introduction to Databases (Spring 2026)
- **DTS110TC** Technology Fundamentals and Perspectives for AI and Big Data Industry (Fall 2025)
- **DTS203TC** Design and Analysis of Algorithms (Spring 2023)
- **DTS205TC** High Performance Computing (Spring 2023)
- **DTS207TC** Database Development and Design (Fall 2023)
- **DTS311TC** Final Year Project (AY2022/23, AY2023/24, AY2024/25, AY2025/26 – Module Leader)
- **DTS407TC** Large Model Techniques (Spring 2025 – Co-teach)
- **ENT001** Explore AI and Entrepreneurship (Fall 2022)

Toyota Motor North InfoTech Lab, Mountain View, CA / NC, USA June 2021 – Sep 2021
Research Scientist Intern

Key Responsibilities:

- Investigated edge computing architectures, vehicular telematics, and over-the-air (OTA) software update algorithms for connected and autonomous vehicles.
- Developed high-efficiency data aggregation models for real-time high-definition (HD) map reconstruction in distributed vehicular networks.

Key Achievements:

- Formulated an adaptive wireless transmission scheduling algorithm optimizing bandwidth and latency constraints for vehicular cloud-edge offloading.
- Co-authored technical research contributions utilized in enterprise telematics research.

University of North Carolina at Charlotte, Charlotte, NC, USA Jan 2016 – May 2022
Graduate Research & Teaching Assistant

Key Responsibilities:

- Conducted doctoral research in wireless edge computing, video streaming optimization, Deep Neural Network (DNN) model compression, and mobile Augmented Reality.
- Delivered recitation and laboratory instruction for undergraduate and graduate networking and electrical engineering courses.

Key Achievements:

- Authored/co-authored 14 peer-reviewed conference and journal publications, including papers in IEEE INFOCOM, IEEE GLOBECOM, IEEE ICC, IEEE SECON, and IEEE Transactions on Network and Service Management.
- Defended doctoral dissertation titled *"Analysis and Enhancement of Resource-hungry Applications"* (Advisor: Prof. Linda Xie).

RESEARCH GRANTS & FUNDED PROJECTS

Total Secured Project & Research Funding: Over RMB 11.2 Million

AI-enabled Optimization for Short Video Streaming Applications Jan 2023 – Dec 2025
Principal Investigator (PI) | Amount: RMB 100,000

- Funding Body: XJTLU Research Development Fund (RDF-22-01-094).
- Status: Successfully Completed (Delivered 7+ papers, 2 patents, edge prototypes).

Benchmarking Framework for In-Car Voice Assistants in Electric Vehicles AY2025 – 2026
Principal Investigator (Co-PI) | Amount: RMB 50,000

- Funding Body: Centre of Excellence for Syntegrative Education (CoESE) Round 2 Grant. Institutional Support.
- Status: Ongoing (EV voice assistant benchmarking protocols and student research).

XJTLU-PAI Joint AI Research Center 2024 – 2029
Co-Principal Investigator (Co-PI) | Amount: RMB 5,000,000

- Funding Body: Enterprise Collaborative Research Fund (Joint Lab).
- Status: Ongoing (3D scene reconstruction, deep learning image stitching, perceptual quality).

Intelligent Toning Process Optimization Project 2024 – 2028
Co-Principal Investigator (Co-PI) | Amount: RMB 4,576,200

- Funding Body: Enterprise Collaborative Research Fund (Taicang Industrial Project).
- Status: Ongoing (Full-cycle industrial machine learning and data mining automation).

Computer-Aided Quantitative Analysis for Digital Pathological Images July 2023 – June 2026
Co-Principal Investigator (Co-PI) | Amount: RMB 500,000

- Funding Body: Governmental Research Grant (Jiangsu Province / Gusu Leading Talents Programme).
- Status: Ongoing (Medical AI diagnostic image analysis).

Suzhou Multimodal Big Data Innovation Application Lab Sep 2023 – Aug 2025
Co-Investigator (Co-I) | Amount: RMB 1,000,000

- Funding Body: Suzhou Municipal Science and Technology Programme.
- Status: Ongoing (Multimodal big data analytics infrastructure).

Video Anomaly Detection Based on Deep Learning Oct 2023 – Dec 2023
Co-Principal Investigator (Co-PI) | Amount: RMB 100,000

- Funding Body: Enterprise Collaborative Research Project with PGsensing Nanjing (Pure ID: 40287877).
- Status: Completed (Object detection frameworks for automated surveillance anomaly detection).

AI Project Design and Implementation Based on Edge Computing Dec 2023 – Nov 2024
Co-Principal Investigator (Co-PI) | Amount: RMB 50,000

- Funding Body: Ministry of Education (MoE) Industry-Academia Collaboration Cooperative Education Program.
- Status: Completed (Edge computing curriculum modules and student project design).

Beyond Boundaries: Syntegrating Capstone, FYP, and Industry Innovations Sep 2024 – Aug 2026
Co-Principal Investigator (Co-PI) | Institutional Support

- Funding Body: Centre of Excellence for Syntegrative Education (CoESE) Internal Grant.
- Status: Ongoing (Syntegrative cross-school Capstone and FYP innovation models).

A Pilot Interdisciplinary Programme – BEng Smart City and Spatial Planning May 2026 – Oct 2026
Co-Investigator (Co-I) | Institutional Support

- Funding Body: XJTLU TDF / EASE-focused Project (EASE-R1-2526-92).
- Status: Ongoing (Cross-school AI and spatial planning curriculum innovation).

Postgraduate Research Scholarship (PGRS) Fee-Only Doctoral Allocation 2024 – Present
Recipient / PhD Supervisor | Full Tuition Waiver Support

- Funding Body: XJTLU Internal Doctoral Research Scholarship.

- Status: Secured (Doctoral student recruitment and supervision).

SCHOLARLY PUBLICATIONS

Citation Summary: 320+ Citations, h-index: 7 (Scopus / Pure Profile)

Part A: Publications Realized Since Joining XJTLU (2022 – Present)

(Denotes supervised undergraduate or postgraduate student co-author)*

[1] Gong, H., Feng, Y.*, Zhang, Z.*, Hou, X., Liu, J., **Huang, S.**, & Liu, H. (2024). **A Dataset and Model for Realistic License Plate Deblurring**. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 776–784). doi:10.24963/ijcai.2024/86.
(CORE A, CCF A, Top-14 Global AI Venue, h5-index: 136. Supervised student co-authors.)*

[2] **Huang, S.**, Xie, J., & Muslam, M. M. A. (2023). **A Cloud Computing Based Deep Compression Framework for UHD Video Delivery**. *IEEE Transactions on Cloud Computing*, 11(2), 1562–1574. doi:10.1109/TCC.2022.3168051.
(SCI, JCR Q1, IF: 6.5, Scopus 16 Citations.)

[3] Zhou, M., Zhu, G., Feng, Z., Lian, H., & **Huang, S.** (2025). **CTLane: An End-to-End Lane Detector by a CNN Transformer and Fusion Decoder for Edge Computing**. *ITU Journal on Future and Evolving Technologies*, 2025(1), 149–161.
(Open Access, 1,300+ Downloads on Pure. Edge-assisted autonomous driving.)

[4] Hu, H.*, & **Huang, S.** (2024). **A Kalman Filtering and Least Absolute Residuals Based Time Series Data Reconstruction Strategy for Structural Health Monitoring**. In *2024 IEEE 10th World Forum on Internet of Things (WF-IoT 2024)* (pp. 1–6). IEEE. doi:10.1109/WF-IoT62078.2024.10811210.
(IEEE Flagship IoT Conference. Co-authored with supervised undergraduate student Haoqi Hu. Corresponding author.)

[5] Hong, Y., Huang, X., Li, F., **Huang, S.**, Weng, Q., Fraidenraich, D., & Voiculescu, I. (2024). **Data-Driven Maturity Level Evaluation for Cardiomyocytes Derived from Human Pluripotent Stem Cells**. *Electronics*, 13(24), 4985. doi:10.3390/electronics13244985.
(Invited Paper, MDPI, SCI, JCR Q2, Open Access.)

[6] Lu, F.*, Tan, J., Li, B.*, & **Huang, S.** (2025). **DDyF-Net: A Depth-Guided Dynamic Filtering Driven Image Dehazing Framework for Dense Haze**. In *Proceedings of the 2025 8th International Conference on Big Data and Artificial Intelligence (BDAI 2025)* (pp. 435–440). IEEE. doi:10.1109/BDAI66031.2025.11325121.
(IEEE International Conference. Co-authored with supervised students Fei Lu and Binyu Li.)

[7] Yuan, H.*, Tan, J., Yan, Y.*, Gao, Z.*, & **Huang, S.** (2025). **Layout-to-Image Driven Data Augmentation for Object Detection**. In *Proceedings of the 2025 8th International Conference on Big Data and Artificial Intelligence (BDAI 2025)* (pp. 303–309). IEEE. doi:10.1109/BDAI66031.2025.11325263.
(IEEE International Conference. Co-authored with supervised students Huajun Yuan, Yuhang Yan, and Zhenkai Gao.)

[8] Tan, J., Gao, Z.*, Yan, Y.*, Zhang, Z.*, & **Huang, S.** (Under Review). **Generative AI Driven Real-time Open-set Object Detection for Mobile AR Applications**. Submitted to *ACM Multimedia Asia 2025*.
(Premier Asian Multimedia Venue. Co-authored with supervised undergraduate students.)

[9] Ye, Z.*, Wang, Y.*, Chen, Q., **Huang, S.**, & Li, H. (Under Review). **Interactive Cyberbullying Detection based on Tree-Structured Hierarchical Attention Networks**. Submitted to *IEEE Transactions*.
(Major NLP and Social Computing Submission. Co-authored with supervised students Ziyu Ye and Yihan Wang.)

[10] **Huang, S.**, et al. (Under Review). **An LLM-Powered Two-Step Framework for Classical Chinese Poetry Translation**. Submitted to *Expert Systems with Applications / Applied Soft Computing*.
(Million-level trilingual corpus and lightweight LLM translation framework. Corresponding author.)

Part B: Prior Publications (Doctoral & Pre-XJTLU Tenure, 2017 – 2022)

[11] Huang, X., Ansari, N., **Huang, S.**, & Li, W. (2022). **Dynamic Bayesian Network Based Security Analysis for Physical Layer Key Extraction**. *IEEE Open Journal of the Communications Society*, 3, 379–390. doi:10.1109/OJCOMS.2022.3154626.
(Invited Paper, IEEE Open Journal, ESCI, JCR Q1, IF: 6.2.)

[12] **Huang, S.**, Huang, X., & Ansari, N. (2021). **Budget-aware Video Crowdsourcing at the Cloud-enhanced Mobile Edge**. *IEEE Transactions on Network and Service Management*, 18(4), 4455–4467. doi:10.1109/TNSM.2021.3102434.
(SCI, JCR Q1, Top Journal in Network Management.)

[13] **Huang, S.**, & Xie, J. (2021). **DAVE: Dynamic Adaptive Video Encoding for Live Video Streaming Applications**. In *Proceedings of the 2021 IEEE International Conference on Sensing, Communication and Networking (SECON)* (pp. 1–9). IEEE.
(IEEE Flagship Sensor and Networking Conference.)

[14] **Huang, S.**, & Xie, J. (2021). **Pearl: A Fast Deep Learning Driven Compression Framework for UHD Video Delivery**. In *Proceedings of the IEEE International Conference on Communications (ICC 2021)* (pp. 1–6). IEEE.
(IEEE Flagship Communications Conference.)

[15] Liang, Z., **Huang, S.**, Huang, X., Cao, R., & Yu, W. (2020). **Post-click Behaviors Enhanced Recommendation System**. In *Proceedings of the 2020 IEEE International Conference on Information Reuse and Integration for Data Science (IRI 2020)* (pp. 267–274). IEEE.
(IEEE Data Science Conference.)

[16] Tapang, D. K., **Huang, S.**, & Huang, X. (2020). **QoE-Based Server Selection for Mobile Video Streaming**. In *Proceedings of the 2020 IEEE/ACM Symposium on Edge Computing (SEC 2020)* (pp. 435–439). IEEE/ACM.
(CCF C, Premier Edge Computing Symposium.)

[17] **Huang, S.**, Han, T., & Xie, J. (2019). **A Smart-Decision System for Realtime Mobile AR Applications**. In *Proceedings of the 2019 IEEE Global Communications Conference (GLOBECOM 2019)* (pp. 1–6). IEEE.
(IEEE Flagship Global Communications Conference.)

[18] Liu, Q., **Huang, S.**, Opadere, J., & Han, T. (2018). **An Edge Network Orchestrator for Mobile Augmented Reality**. In *Proceedings of the 2018 IEEE Conference on Computer Communications (INFOCOM 2018)* (pp. 1–9). IEEE.
(CORE A, CCF A, Premier Networking Conference.)*

[19] **Huang, S.**, Liu, Q., Han, T., & Ansari, N. (2017). **Data-Driven Network Optimization in Ultra-Dense Radio Access Networks**. In *Proceedings of the 2017 IEEE Global Communications Conference (GLOBECOM 2017)* (pp. 1–6). IEEE.
(IEEE Flagship Global Communications Conference.)

[20] Liu, Q., **Huang, S.**, Deng, Y., & Han, T. (2017). **Demo Abstract: MExR: Mobile Edge Resource Management for Mixed Reality Applications**. In *Proceedings of the 2017 IEEE INFOCOM Workshops* (pp. 990–991). IEEE.
(IEEE INFOCOM Workshop Demo.)

[21] **Huang, S.**, Han, T., & Ansari, N. (2017). **Big-Data-Driven Network Partitioning for Ultra-Dense Radio Access Networks**. In *Proceedings of the 2017 IEEE International Conference on Communications (ICC 2017)* (pp. 1–6). IEEE.
(IEEE Flagship Communications Conference.)

AWARDS AND HONORS

- **Second Prize of Annual Curriculum Innovation (Team Leader):** The First XJTLU Innovative Cases of Industry-Education Integrated Curriculum Competition, 2024.
- **Annual Performance Rating – OUTSTANDING:** XJTLU Professional Development Review (PDR), AY2024/25.
- **Annual Performance Rating – OUTSTANDING:** XJTLU Professional Development Review (PDR), AY2023/24.
- **Annual Performance Rating – EXCELLENT:** XJTLU Professional Development Review (PDR), AY2022/23.
- **National Second Prize & Regional First Prize Supervisor:** 2023 Global Campus AI Algorithm Elite Competition.
- **Regional Second Prize Supervisor:** National Student Smart Car Competition.
- **Bronze Medal Supervisor:** "CP Cup" 2024 Global College Student Employment and Entrepreneurship Competition.
- **Second Prize Supervisor:** Global Environment Solutions Challenge, 2025.
- **School Honored Student Award:** School of Software Engineering, Sun Yat-sen University, 2015.

PATENTS & INTELLECTUAL PROPERTY

- **Authorized Patent:** Hongbin Liu, Haoyan Gong, & **Siqi Huang**. *A Generative License Plate Image Deblurring Method Based on Multi-Scale Inputs.* IP No. Pat2024-0018 (Authorized).
- **Patent Application: Siqi Huang**, Jiexi Tan, Zhenkai Gao, Yuhang Yan, & Zimu Zhang. *A Real-Time Open-Set Object Detection System and Method for Mobile Augmented Reality Devices.* IP Application No. Pat2025-0042 (Filed).

ADMINISTRATIVE & ACADEMIC SERVICE AT XJTLU

Programme Director, BEng Data Science and Big Data Technology (DSBD) 2023 – Present

- **Responsibilities:** Provide comprehensive academic leadership for one of XJTLU's flagship and largest undergraduate cohorts; oversee curriculum review, teaching allocations, assessment consistency, and student academic progression; lead Liverpool (UoL) exchange mapping.
- **Key Achievements:** Successfully led and gained institutional approval for a major full-cycle programme modification incorporating Syntegrative Education; established formal module mapping for student study abroad at UoL; sustained high cohort progression and positive student feedback.

Student Recruitment and Outreach Officer, School of AIAC 2022 – Present

- **Responsibilities:** Plan, coordinate, and execute recruitment and dissemination strategies targeting prospective Year-1 undergraduate students and parents; liaise with University Admissions and Marketing teams.
- **Key Achievements:** Organized and personally led more than 50 major recruitment sessions, campus open days, high school visits, and marketing video interviews; consistently met and exceeded school undergraduate intake targets each academic year; mentored incoming colleagues in recruitment outreach.

Founder & Director, AIAC Metaverse Research Lab 2023 – Present

- **Responsibilities:** Established laboratory computing infrastructure (studio displays, high-capacity Synology NAS, edge AI devices); supervise student research teams; manage internal RDF and enterprise collaborative project experiments.
- **Key Achievements:** Mentored over 10 students across 7 research projects and 5 SURF summer projects; generated 7+ peer-reviewed publications and 2 patents directly from lab research.

Member, XEC UG Curriculum Review Task Force / College Curriculum Review Panel 2024 – Present

- **Responsibilities:** Review and evaluate undergraduate programme modifications and new module proposals across XEC schools; ensure alignment with Syntegrative Education standards and University of Liverpool regulations.

- **Key Achievements:** Contributed to enhancing curriculum coherence across multiple College programmes, fostering interdisciplinary synergies and industry-engaged projects.

AIAC Pathfinder & 2+1+1 / 2+2 Overseas Programme Coordinator 2023 – Present

- **Responsibilities:** Collaborate with the Entrepreneurship and Enterprise Hub (EEH) to guide student innovative ventures; manage academic advising and credit transfer for students participating in international exchange.
- **Key Achievements:** Facilitated seamless academic transitions for study abroad students; supported student startup teams entering X3 CoVenture and external competitions.

University Research Representative for School of AIAC 2023 – Present

- **Responsibilities:** Represent the School on university-level research liaison committees; disseminate research policy updates, grant calls, and Pure research intelligence.
- **Key Achievements:** Enhanced school research visibility and facilitated cross-school interdisciplinary collaborations in smart cities and digital healthcare.

Internal Reviewer for University Grant & Progression Panels 2023 – Present

- **Responsibilities:** Review proposals submitted to the Teaching Development Fund (TDF) and Postgraduate Research Scholarship (PGRS) panels; assess Ph.D. annual progression milestones.
- **Key Achievements:** Delivered rigorous, constructive peer reviews ensuring institutional funding excellence and academic standards.

Innovation & Entrepreneurship Mentor, Taicang University Science & Technology Park 2023 – Present

- **Responsibilities:** Advise student startups and enterprise incubators; cultivate enterprise-university research partnerships with local industries in Taicang.
- **Key Achievements:** Brokered an enterprise partnership with an autonomous vehicle company resulting in the donation of two test vehicles (valued at RMB 300,000) for school research.

EXTERNAL ACADEMIC SERVICE & PROFESSIONAL MEMBERSHIPS

Professional Memberships:

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, IEEE Communications Society
- Member, IEEE Computer Society

Peer Reviewer for Premier International Journals:

- IEEE Transactions on Cloud Computing (TCC)
- IEEE Transactions on Network and Service Management (TNSM)
- IEEE Open Journal of the Communications Society
- Electronics (MDPI)
- Sensors (MDPI)

Technical Program Committee (TPC) & Conference Reviewer:

- IEEE International Conference on Computer Communications (INFOCOM)
- IEEE Global Communications Conference (GLOBECOM)
- IEEE International Conference on Communications (ICC)
- IEEE Conference on Sensing, Communication and Networking (SECON)
- IEEE World Forum on Internet of Things (WF-IoT)
- International Conference on Big Data and Artificial Intelligence (BDAI)